

Simplified Attendance Mobile App using Realm Database

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Technologies used :

Frontend : Java (Andriod studio)

Backend : Realm Database (Exclusive Mongo database for mobile applications)

Introduction:

The Simplified Attendance App is a user-friendly application designed to streamline the process of taking attendance in the classroom. This project report outlines the development and implementation of the app using Java in Android Studio and RealmDB for local data storage. The report will cover the app's functionality, design, technology stack, and provide sample code for establishing a database connection.

App Functionality:

The Simplified Attendance App aims to simplify the attendance-taking process for teachers. Its key features include:

- a. Class Creation: Teachers can create a class by providing a class name, subject name, and selecting a theme for the class interface.
- b. Student Management: Teachers can add students to each class, enabling them to maintain a comprehensive list of students.
- c. Attendance Tracking: Teachers can quickly take attendance for each class and submit it with minimal effort.
- d. Report Generation: The app generates attendance reports for each class, allowing teachers to view and analyze attendance data easily.

App Design

The app boasts a clean and elegant user interface (UI) that is tailored to the needs of teachers. The design elements prioritize simplicity and intuitive navigation, ensuring a smooth user experience. The UI components and layout have been carefully crafted to enhance readability and minimize distractions.

Technology Stack

The Simplified Attendance App has been developed using Java programming language and Android Studio as the development environment. The primary technology utilized in this project is RealmDB, a local database designed specifically for mobile applications. RealmDB offers efficient data storage, easy integration, and high-performance queries.

To integrate Backend RealmDB plugin into the database :

build.gradle (Application)

```
// Top-level build file where you can add configuration options common to  
all sub-projects/modules.  
buildscript {  
    repositories {  
        google()  
        jcenter()  
    }  
    dependencies {  
        classpath "com.android.tools.build:gradle:4.0.2"  
        classpath "io.realm:realm-gradle-plugin:6.0.2"  
  
        // in the individual module build.gradle files  
    }  
}  
allprojects {  
    repositories {  
        google()  
        jcenter()  
        maven { url "https://jitpack.io" }  
    }  
}  
task clean(type: Delete) {  
    delete rootProject.buildDir }  
}
```

Implementation Details

a. Project Setup:

To create the project in Android Studio and set up the necessary dependencies, follow these steps:













